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Transportable 1555-nm Ultra-Stable Laser with Sub-0.185-Hz Linewidth *

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We present two cavity-stabilized lasers at 1555 nm, which are built to be the frequency source for a transportable photonic microwave generation system. The frequency instability reaches the thermal noise limit (7×10^{-16}) of the 10-cm ultra-low expansion glass cavity at 1–10 s averaging time and the beat signal of the two lasers reveals a remarkable linewidth of 185 mHz.

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Ultra-stable lasers have become key elements for accurate modern measurements, including optical atomic clock,^[1] gravitational wave detection,^[2] fundamental physics tests,^[3] and ultra-low phase noise microwave generation.^[4] An ultra-stable laser with a wavelength of 1.5 μm ^[5–7] at the transmission window of fiber communication enables long distance ultra-stable frequency transfer over fiber links.^[8,9] Combined with the optical frequency comb, the frequency stability of these lasers could be extended to other optical wavelengths and into the microwave domain.^[10] The frequency synthesis from a light wave to a microwave does not decay the stability of the laser. Therefore, it provides a new way to produce ultra-stable microwave signals for applications such as in cesium fountain clocks.^[11,12]

The high frequency stability of the ultra-stable laser is achieved by locking the continuous wave lasers to the optical length of the ultra-stable Fabry–Pérot (FP) cavities. In state-of-the-art systems,^[5–7,13] the fundamental limitation of laser frequency stability is the thermal fluctuation of cavity length, which is primarily caused by the Brownian motion of the mirrors and coatings. This effect could be further minimized by extending the cavity length,^[14] enlarging the beam size of the cavity,^[15] manufacturing the cavity using materials with lower mechanical loss,^[16] or placing the cavity in a cryogenic environment.^[5] With a 210-mm silicon single-crystal cavity operated at 124 K, a fractional frequency instability of 1×10^{-16} at short timescales and a laser linewidth below 40 mHz at 1.5 μm are obtained.^[5]

Here we report 1555-nm ultra-stable lasers for simple and transportable applications. The beatnote between the two independent lasers exhibits a remark-

able linewidth of 0.185 Hz. The relative frequency instability of these lasers reaches the thermal noise limitation of 7×10^{-16} within 1–10 s averaging time. The thermal-limited performance of the ultra-stable laser system provides a good foundation to realize a low noise 9.2 GHz local oscillator for the cesium fountain clock lab of our institute.

To minimize acceleration sensitivity, an ultra-stable cavity is designed with the ideas of symmetry and balance. Our cavity is a 100 mm-long, 110-mm-diameter cylinder with a 7-mm-thick, 140-mm-diameter flange on the mid-plane. All sizes are calculated by finite element modeling (FEM), and we find an acceleration sensitivity of $1.1 \times 10^{-10}/\text{g}$ along the cavity length axis. The cavity is vertically and rigidly mounted, which causes the cavity length to be insensitive vertical acceleration in (along) the direction of gravity. Both the cavity and the mirror substrate are made of ultra-low expansion glass (ULE) glass, which has a very low coefficient of thermal expansion (CTE), on the level of about 10^{-8} . The cavity is located between two layers of thermal shield, which is made of gold-plated copper, and a dual cantilever is applied to hold the cavity at three equally distributed points on the flange.^[6] The cantilever is made of aviation aluminum, which has good mechanical properties and is light weight, and the cantilever is attached to an invar support frame to increase the holding strength and to reduce the thermal expansion difference between the cavity and the inner thermal shield, consequently minimizing stress and deformation of the cavity. The outermost layer is the vacuum chamber, which has a vacuum level below 4×10^{-7} Torr. All components are bolted together with elastic washers for damping vibration, and ceramic sleeves are used for thermal

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insulation.

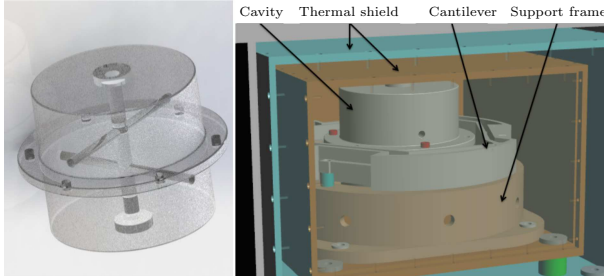


Fig. 1. Geometry of the cavity and assembly of the cavity in vacuum chamber.

The thermal noise, introduced by the Brownian motion of the mirror substrates and coatings, fundamentally limits the length instability of the cavity, and the corresponding double-sideband power spectrum of the mirror displacement is given by^[17]

$$G_{\text{mirror}}(f) = \frac{4k_B T}{\omega} \frac{1 - \sigma^2}{\sqrt{\pi} E w_0} \left(\varphi_{\text{sub}} + \frac{2}{\sqrt{\pi}} \frac{1 - 2\sigma}{1 - \sigma} \frac{d}{w_0} \varphi_{\text{coat}} \right), \quad (1)$$

where k_B is the Boltzmann constant, T is the temperature, and $\omega = 2\pi f$ is the angular frequency. These terms constitute the transfer function from stress to displacement, and σ is Poisson's ratio, E is Young's modulus (67.6 GPa), φ_{sub} is the mechanical loss of the mirror substrate (1.67×10^{-5}), w_0 is the beam radius (385 μm), d is the coating thickness (8.4 μm), and φ_{coat} is the mechanical loss of coating (4×10^{-4}). The first and second terms in parentheses represent the contribution from the mirror substrate and coating, respectively. Equation (1) gives the displacement spectrum of one mirror, which should be multiplied by two, since two mirrors are included. As mentioned before, the cavity is 10-cm long and constructed of ULE glass, and a pair of ULE mirrors (which construct a plano-concave geometry with a 1-m radius of curvature) is used to enlarge the beam size as much as possible. Since the relative frequency instability of the laser, $\Delta f/f$, is determined by the cavity length, $\Delta L/L$, the double-sideband thermal frequency noise spectrum, $S_\nu(f)$, is estimated to be $0.01 \times (1 \text{ Hz}/f) \text{ Hz}^2/\text{Hz}$, and the Allan deviation is 6.2×10^{-16} . Generally speaking, the contribution of thermal elastic noise caused by the thermal expansion of the mirror substrates heated by the laser beam spot should be considered.^[18] However, for mirror substrates constructed with ULE glass, such as in this case, the contribution is negligible (0.01%). In addition, previous studies have shown that the thermal elastic effect is much stronger in the coating than in the substrate due to larger CTE.^[19] In the present case, due to the small thickness and low CTE of ULE

substrates, the thermal elastic noise of the coatings is also negligible (approximately 0.5%).

Finally, to reduce the noise contribution introduced by mechanical vibration, acoustic noise, airflow, and temperature fluctuations, the free-space optics are located on a vibration isolation platform, and they are shielded by an acoustic isolation enclosure made of a metal framework covered with polyester fiberboard and acoustic absorption foam. Note that the laser is put out of the environmental protection for convenience.

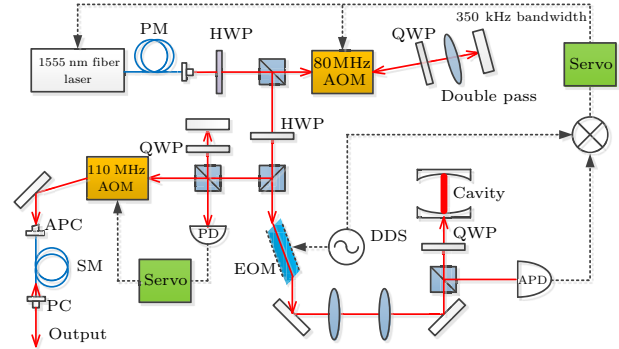


Fig. 2. Schematic diagram of the experimental setup. Red line: free space optical path; blue line: fiber optical path; and black dashed line: electric path. PM: polarization maintaining fiber; HWP: half-wave plate; QWP: quarter-wave plate; AOM: acousto-optic modulator; EOM: electro-optic modulator; DDS: direct digital synthesizer; SM: single mode fiber; PC: physical contact; APC: angled physical contact; PD: photodiode; and APD: avalanche photodiode.

Figure 2 demonstrates the experimental setup of an ultra-stable laser, whose frequency is stabilized using the well-known Pound–Drever–Hall (PDH) technique.^[20] Here we constructed two independent systems with the same configuration for performance evaluations. A commercial fiber laser at 1555 nm (NKT Photonics Koheras Adjustik-E15) is locked to the reference cavity with a finesse of 610000 (310000 for the other system) that is measured by the cavity ring down method.^[21] To obtain broadband frequency control, the laser passes twice an acousto-optic modulator (AOM), which shifts the laser frequency by approximately 80 MHz for each pass. The AOM is driven by a voltage controlled oscillator (VCO), which provides a 0.35 MHz servo bandwidth for laser frequency control. The double-pass configuration enables a broad tuning range of ~ 10 MHz (full width of half maximum power) and compensates for beam deflections.^[22] Then, the laser is modulated by a free space electro-optic modulator (EOM) cut at Brewster's angle, which has an ultra-low residual amplitude modulation (RAM) level of a few parts per million (ppm), corresponding to frequency instability at the 10^{-17} level.^[23] In this case, about 8 μW modulated light, with a modulation depth of ~ 0.7 rad, is coupled

to the cavity. The transmission light is $\sim 1.5 \mu\text{W}$ for both systems when the laser is frequency-locked onto the TEM00 mode of the cavity. The frequency sensitivity to the transmission power is $10 \text{ Hz}/\mu\text{W}$ (for the 610000 finesse cavity) and $5 \text{ Hz}/\mu\text{W}$ (for the 310000 finesse cavity). Therefore, the frequency drift is less than 0.15 Hz for both cavities, while the optical power fluctuation is below 1% . No extra power stabilization is required, since the power of the commercial laser is quite stable (well below 1% per second). The cavity-reflected light ($\sim 5 \mu\text{W}$) is directed to an avalanche photodetector (APD), yielding the error signal after being demodulated. Finally, the loop filter converts the error signal from the negative-feedback control signals to the piezoelectric transducer (PZT) of the fiber laser for large range control and the VCO for fast noise correction.

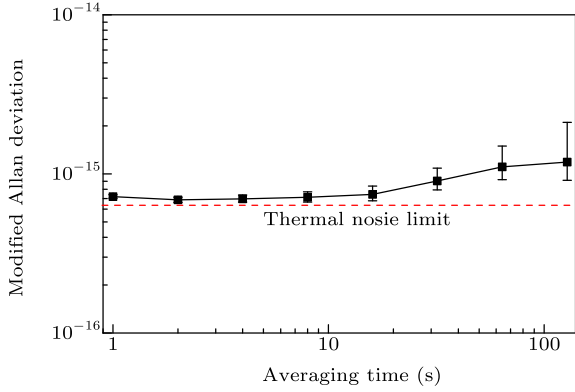


Fig. 3. Fractional frequency instability of one stabilized laser. Black dashed line: thermal-noise-limited frequency instability, 6.2×10^{-16} .

To evaluate the performance of the ultra-stable laser, we compare two identical systems located in two different labs that were linked by an 80-m-long SMF-28 fiber. Using the fiber noise cancellation technique,^[24] we transfer one laser to another and then combine both lasers in a photodetector. The beatnote signal is recorded by a dead time free Π -type counter (K+K Messtechnik, model FXQE80) in phase averaging mode, and the modified Allan deviation is employed to indicate frequency instability.^[5,25] Figure 3 demonstrates the relative frequency instability of one laser ($\sim 1 \text{ Hz/s}$ linear drift removed, usually this value is below 0.3 Hz/s) by dividing the normalized deviation by $\sqrt{2}$, since two systems should have the same noise level. The frequency instability is about 7×10^{-16} from 1 to 10 s averaging time, which is close to the calculated thermal noise limit of 6.2×10^{-16} . The instability rises to 1×10^{-15} at 100 s, and we attribute this to temperature fluctuations, as the two laser systems are located in two different labs without active thermal control.

The phase noise spectrum $S_\varphi(f)$ of the laser is

shown in Fig. 4. We convert the beatnote between the two lasers into a voltage fluctuation using a home-made frequency-to-voltage converter and analyze the noise level with a fast Fourier transform (FFT) analyzer (SR 785, Stanford). The phase noise of the free running laser is suppressed by more than 50 dB for low frequencies. The right axis is the corresponding phase time noise of the laser, and the time jitter from 1 Hz to 100 kHz is integrated to be approximately 360 attoseconds.

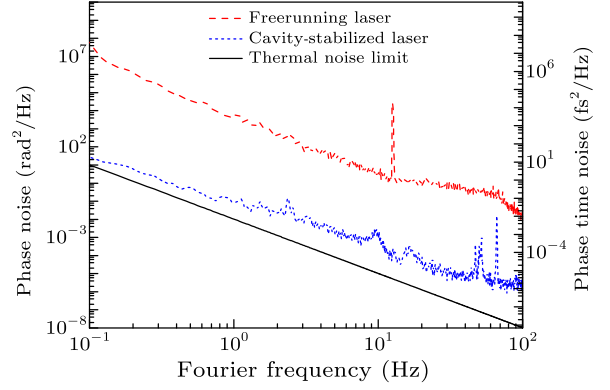


Fig. 4. Phase and time noise power spectral density of the laser. Blue line: phase noise of a stabilized laser; red line: phase noise of a free running laser; and black line: thermal noise limit.

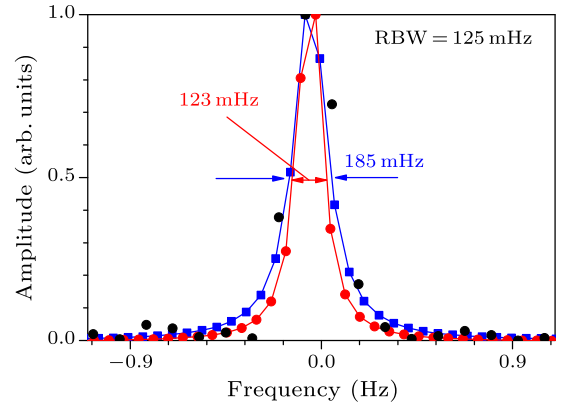


Fig. 5. Linewidth of beatnote between two stabilized lasers. Black dots: beatnote of two stabilized lasers, blue thick line: Lorentz fit of two stabilized lasers, red thin line: Lorentz fit of the beatnote of two microwave signals at the same RBW of 125 mHz .

Figure 5 demonstrates the linewidth of the laser beatnote ($\sim 1 \text{ mHz/s}$ drift) measured with an FFT analyzer at a resolution bandwidth (RBW) of 125 mHz . From the fit with the Lorentz line shape, the full width at half maximum (FWHM) is 185 mHz , which is rather narrow considering the use of the 10-cm-long entire ULE cavity.^[5,7] The measurement noise floor (123 mHz) is obtained by measuring the beatnote of two stable RF signals under the same RBW, and this result is consistent with the setting value of 125 mHz .

In conclusion, we have reported a cavity-stabilized

ultra-stable laser at 1555 nm based on a commercially available cavity made entirely of ULE. Frequency instability is measured to be 7×10^{-16} from 1 to 10 s. The beatnote between the two systems exhibits a linewidth of about 0.185 Hz, and the real linewidth of one laser should be narrower as a function of the noise contribution of the second laser and the measurement system. Both laser systems are built in our laboratory, and due to the good structural stability, one has been successfully moved to the cesium fountain clock lab to realize ultra-low phase noise 9.2 GHz microwave generation as the local oscillator. Since the frequency instability of the laser is well below that of the best fountain clocks, we can conclude that the laser system is competent for this task.

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